

Hyperbolas

ANSWER KEY



Standard form

- 1 For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:
 - a the vertices $(\pm 4, 0)$.
 - b the foci $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.
 - c the equations of the asymptotes $y = \pm \frac{3}{4}x$.
 - 2 Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.
 $a = 2, c = 3, b^2 = 9 - 4 = 5: \frac{y^2}{4} - \frac{x^2}{5} = 1$.
 - 3 Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.
 - a Write the equation in standard form by completing the square. $4(x + 1)^2 - (y - 2)^2 = 4$, i.e. $(x + 1)^2 - \frac{(y - 2)^2}{4} = 1$.
 - b Identify the conic and state its center. A hyperbola opening left-right with center $(-1, 2)$.
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Identifying conics

- 4 Identify each conic section from its equation:
 - a $x^2 + y^2 = 49$ Circle.
 - b $\frac{x^2}{9} + \frac{y^2}{25} = 1$ Ellipse.
 - c $y^2 - 4x^2 = 16$ Hyperbola.
 - d $y = 3x^2 - 6x + 1$ Parabola.
 - 5 A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.
 $a = 3$ and $\frac{b}{a} = 2$ gives $b = 6: \frac{x^2}{9} - \frac{y^2}{36} = 1$.
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Hyperbolas centered away from the origin

- 6 For the hyperbola $\frac{(y + 1)^2}{4} - \frac{(x - 3)^2}{9} = 1$, find:
 - a the center $(3, -1)$
 - b the vertices $(3, -1 \pm 2) = (3, 1), (3, -3)$
 - c the direction of opening opens up/down
 - 7 Complete the square to write $9x^2 - 4y^2 - 18x - 16y - 43 = 0$ in standard form, and state the center.
 $9(x^2 - 2x) - 4(y^2 + 4y) = 43: 9(x - 1)^2 - 9 - 4(y + 2)^2 + 16 = 43$, so $9(x - 1)^2 - 4(y + 2)^2 = 36$, i.e. $\frac{(x - 1)^2}{4} - \frac{(y + 2)^2}{9} = 1$. Center: $(1, -2)$.
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Eccentricity and applications

- 8 Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

$c = \sqrt{9 + 16} = 5$; $e = \frac{5}{3} \approx 1.67$. Since $e > 1$ and relatively large, the branches are wide and open quickly, resembling straight lines more than a tight curve.

- 9 A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

$2a = 120$, so $a = 60$; $2c = 200$, so $c = 100$; $b^2 = c^2 - a^2 = 10000 - 3600 = 6400$. Equation:

$$\frac{x^2}{3600} - \frac{y^2}{6400} = 1.$$